

WORKED EXAMPLE

Turn inequalities into neuron tests

A half-space is the set of points on one side of a line, including the line here.

Study region: the closed rectangle $1 \leq x \leq 3$ and $0 \leq y \leq 2$. We need all four conditions to hold.

$$x \geq 1 \Rightarrow x - 1 \geq 0 \Rightarrow h_1 = H(x - 1)$$

$$x \leq 3 \Rightarrow 3 - x \geq 0 \Rightarrow h_2 = H(3 - x)$$

$$y \geq 0 \Rightarrow y \geq 0 \Rightarrow h_3 = H(y)$$

$$y \leq 2 \Rightarrow 2 - y \geq 0 \Rightarrow h_4 = H(2 - y)$$

$H(s) = 1$ when $s \geq 0$, else 0. This applies the half-space construction in Lecture 2, page 100.

WORKED EXAMPLE

Require all four tests at the output

Lecture 2, page 108 combines half-space decisions using AND.

$$R_1 = H(h_1 + h_2 + h_3 + h_4 - 3.5)$$

$$\text{all four true : } R_1 = H(4 - 3.5) = H(0.5) = 1$$

$$\text{at most three true : } \sum_j h_j - 3.5 \leq -0.5 \Rightarrow R_1 = 0$$

The four hidden neurons turn geometric tests into bits. The output checks those bits, not the original x and y values.

Their weights for (x, y) and biases are: $(1, 0), -1$; $(-1, 0), 3$; $(0, 1), 0$; $(0, -1), 2$.

Because equality passes each test, the rectangle includes its edges and corners.

WORKED EXAMPLE

Inside: point (2, 1)

Evaluate every half-space, then combine their Boolean outputs.

$$h_1 = H(2 - 1) = H(1) = 1, \quad h_2 = H(3 - 2) = H(1) = 1$$

$$h_3 = H(1) = 1, \quad h_4 = H(2 - 1) = H(1) = 1$$

$$R_1 = H(1 + 1 + 1 + 1 - 3.5) = H(0.5) = 1$$

A single failed boundary test rejects the point. Scores equal to zero pass because $H(0) = 1$.

WORKED EXAMPLE

Left of the region: point (0, 1)

Evaluate every half-space, then combine their Boolean outputs.

$$h_1 = H(0 - 1) = H(-1) = 0, \quad h_2 = H(3 - 0) = H(3) = 1$$

$$h_3 = H(1) = 1, \quad h_4 = H(2 - 1) = H(1) = 1$$

$$R_1 = H(0 + 1 + 1 + 1 - 3.5) = H(-0.5) = 0$$

A single failed boundary test rejects the point. Scores equal to zero pass because $H(0) = 1$.

WORKED EXAMPLE

Above the region: point (2, 3)

Evaluate every half-space, then combine their Boolean outputs.

$$h_1 = H(2 - 1) = H(1) = 1, \quad h_2 = H(3 - 2) = H(1) = 1$$

$$h_3 = H(3) = 1, \quad h_4 = H(2 - 3) = H(-1) = 0$$

$$R_1 = H(1 + 1 + 1 + 0 - 3.5) = H(-0.5) = 0$$

A single failed boundary test rejects the point. Scores equal to zero pass because $H(0) = 1$.

WORKED EXAMPLE

At a corner: point (1, 0)

Evaluate every half-space, then combine their Boolean outputs.

$$h_1 = H(1 - 1) = H(0) = 1, \quad h_2 = H(3 - 1) = H(2) = 1$$

$$h_3 = H(0) = 1, \quad h_4 = H(2 - 0) = H(2) = 1$$

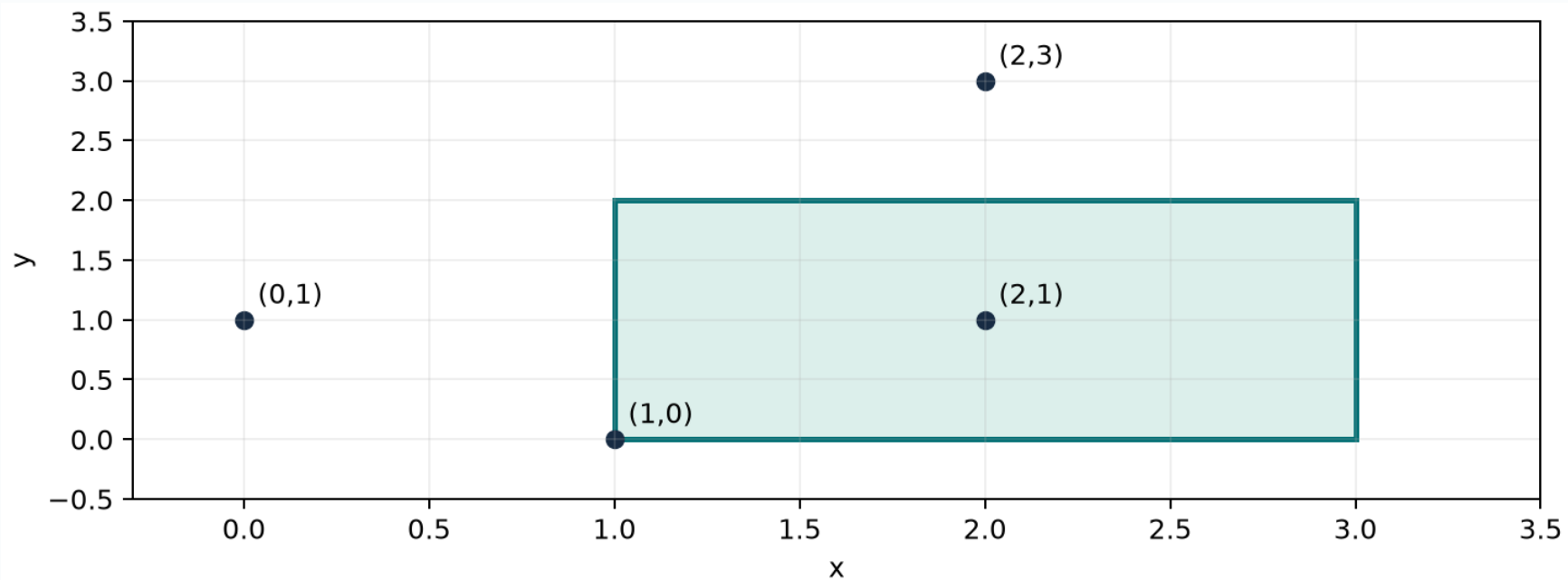
$$R_1 = H(1 + 1 + 1 + 1 - 3.5) = H(0.5) = 1$$

A single failed boundary test rejects the point. Scores equal to zero pass because $H(0) = 1$.

WORKED EXAMPLE

Picture the rectangle and its test points

The shaded region includes all four edges.



The corner (1, 0) passes both equality tests. The points (0, 1) and (2, 3) fail a different half-space each.

WORKED EXAMPLE

Combine separate regions using OR

Lecture 2, page 109: accept a point if either region accepts it.

$$R_2 = H(H(x - 5) + H(6 - x) + H(y) + H(2 - y) - 3.5)$$

R2 is the closed rectangle $5 \leq x \leq 6$, $0 \leq y \leq 2$.

$$U = H(R_1 + R_2 - 0.5)$$

point	R1 hidden bits	R2 hidden bits	(R1,R2)	U
(2,1)	1111	0111	(1,0)	1
(5.5,1)	1011	1111	(0,1)	1
(4,1)	1011	0111	(0,0)	0

For (4,1): $R_1 = H(3 - 3.5) = 0$ and $R_2 = H(3 - 3.5) = 0$. $U = H(0 + 0 - 0.5) = 0$. The gap stays outside.

WORKED EXAMPLE

A pulse is the difference of two steps

Lecture 2, page 115 uses two threshold neurons and a summing output.

$$P(x) = H(x - a) - H(x - b), \quad a < b$$

$$x < a: \quad P = 0 - 0 = 0$$

$$a \leq x < b: \quad P = 1 - 0 = 1$$

$$x \geq b: \quad P = 1 - 1 = 0$$

The interval $[a, b)$ includes a but excludes b . At b , the second step turns on and cancels the first.

This differs from our closed rectangle: pulse subtraction intentionally switches the output off at the right endpoint.

WORKED EXAMPLE

Add two scaled pulses

Our study function has height 2 on $[0,1)$ and height 0.5 on $[1,3)$.

$$f(x) = 2[H(x) - H(x - 1)] + 0.5[H(x - 1) - H(x - 3)]$$

$$= 2H(x) + (-2 + 0.5)H(x - 1) - 0.5H(x - 3)$$

$$= 2H(x) - 1.5H(x - 1) - 0.5H(x - 3)$$

Hidden outputs are $H(x)$, $H(x-1)$, $H(x-3)$. Final weights are (2, -1.5, -0.5) and bias 0.

The final output is a LINEAR sum, not another threshold. Thresholding it would lose the requested heights.

WORKED EXAMPLE

Calculate pulse values: -0.5, 0, 0.5

First threshold the hidden scores, then take the unthresholded weighted sum.

$$x = -0.5 : \quad h = (H(-0.5), H(-1.5), H(-3.5)) = (0, 0, 0)$$

$$f = 2(0) - 1.5(0) - 0.5(0) = 0.0$$

$$x = 0 : \quad h = (H(0), H(-1), H(-3)) = (1, 0, 0)$$

$$f = 2(1) - 1.5(0) - 0.5(0) = 2.0$$

$$x = 0.5 : \quad h = (H(0.5), H(-0.5), H(-2.5)) = (1, 0, 0)$$

$$f = 2(1) - 1.5(0) - 0.5(0) = 2.0$$

WORKED EXAMPLE

Calculate pulse values: 1, 2, 3

First threshold the hidden scores, then take the unthresholded weighted sum.

$$x = 1: \quad h = (H(1), H(0), H(-2)) = (1, 1, 0)$$

$$f = 2(1) - 1.5(1) - 0.5(0) = 0.5$$

$$x = 2: \quad h = (H(2), H(1), H(-1)) = (1, 1, 0)$$

$$f = 2(1) - 1.5(1) - 0.5(0) = 0.5$$

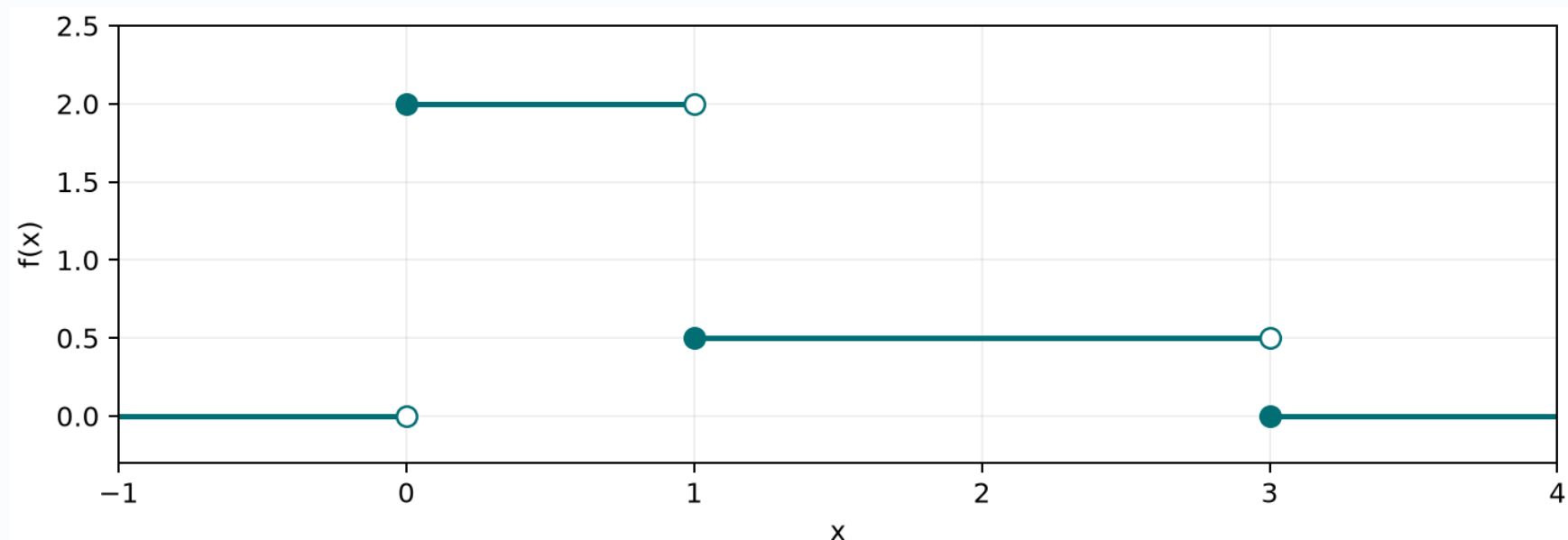
$$x = 3: \quad h = (H(3), H(2), H(0)) = (1, 1, 1)$$

$$f = 2(1) - 1.5(1) - 0.5(1) = 0.0$$

WORKED EXAMPLE

Read the endpoints on the graph

Filled circles are included; open circles are excluded at each jump.



At $x = 1$ the first pulse ends and the second begins: $f(1) = 0.5$. At $x = 3$ both contributions cancel: $f(3) = 0$.

Narrow, scaled pulses are building blocks for approximation (page 116). This finite example is not a proof that one fixed small network exactly fits every function.

INDEPENDENT PRACTICE

Your turn: shift the region and pulses

Use the same $H(0) = 1$ convention.

A. Build the closed rectangle $-1 \leq x \leq 1$ and $2 \leq y \leq 4$. Write its four hidden tests and AND output. Classify $(-1,2)$, $(0,3)$, $(2,3)$.

B. Build a function of height 1.5 on $[-1,0)$ and height 3 on $[0,2)$, zero elsewhere. Give the three hidden steps, output weights and bias.

Calculate the function at $x = -1, 0, 1$ and 2 . Show the hidden outputs before the weighted sum.

Do not threshold the final function-approximation output.

WORKED ANSWER

Practice answer: the shifted rectangle

Translate each lower and upper bound separately.

$$h = (H(x + 1), H(1 - x), H(y - 2), H(4 - y))$$

$$R = H(h_1 + h_2 + h_3 + h_4 - 3.5)$$

$$(-1, 2): \quad h = (H(0), H(2), H(0), H(2)) = (1, 1, 1, 1) \Rightarrow R = 1$$

$$(0, 3): \quad h = (H(1), H(1), H(1), H(1)) = (1, 1, 1, 1) \Rightarrow R = 1$$

$$(2, 3): \quad h = (H(3), H(-1), H(1), H(1)) = (1, 0, 1, 1) \Rightarrow R = 0$$

The first point is an included corner. The last point fails the $x \leq 1$ test.

WORKED ANSWER

Practice answer: derive the pulse weights

Combine the repeated step at $x = 0$ before reading off the weights.

$$f = 1.5[H(x+1) - H(x)] + 3[H(x) - H(x-2)]$$

$$= 1.5H(x+1) + 1.5H(x) - 3H(x-2)$$

Hidden steps: $H(x+1)$, $H(x)$, $H(x-2)$. Weights: (1.5, 1.5, -3), bias 0.

$$x = -1 : h = (1, 0, 0) \Rightarrow f = 1.5(1) + 1.5(0) - 3(0) = 1.5$$

$$x = 0, 1 : h = (1, 1, 0) \Rightarrow f = 1.5(1) + 1.5(1) - 3(0) = 3$$

$$x = 2 : h = (1, 1, 1) \Rightarrow f = 1.5 + 1.5 - 3 = 0$$